

GBA TheoChem Summer School Problem Set

One Comprehensive Problem per Topic

Topics:

- Hartree–Fock Theory (Problem 1)
- Density Functional Theory (Problem 2)
- Configuration Interaction and Coupled-Cluster Theory (Problem 3)
- Excited States (Problem 4)
- Quantum Algorithm Basics for Quantum Chemistry (Problem 5)

Topic 1: Hartree–Fock Theory

Problem 1: Hartree–Fock Energy, Orbitals, and Self-Consistency

These parts connect determinant energy, finite-basis orbital equations, SCF convergence, symmetry breaking, and orbital stationarity.

1. Consider a closed-shell two-electron system in which both electrons occupy a normalized spatial orbital ϕ_1 , with occupied spin orbitals $\phi_1\alpha$ and $\phi_1\beta$. Starting from the spin-orbital Slater–Condon expression, derive

$$E_{\text{el}} = 2h_{11} + (11|11), \quad (1)$$

and show explicitly why the opposite-spin exchange matrix element vanishes.

2. For the symmetric two-AO generalized eigenproblem

$$\mathbf{F}\mathbf{c} = \epsilon\mathbf{S}\mathbf{c}, \quad \mathbf{S} = \begin{bmatrix} 1 & s \\ s & 1 \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} f & g \\ g & f \end{bmatrix}, \quad |s| < 1, \quad (2)$$

derive the $\mathbf{S}$ -normalized symmetric and antisymmetric eigenvectors and their orbital energies.

3. At a closed-shell SCF fixed point, let

$$\mathbf{P} = 2\mathbf{C}_o\mathbf{C}_o^T, \quad \mathbf{F}\mathbf{C}_o = \mathbf{S}\mathbf{C}_o\epsilon_o, \quad (3)$$

where $\mathbf{F}$ and $\mathbf{S}$ are real symmetric matrices. Prove that the nonorthogonal-basis residual

$$\mathbf{R} = \mathbf{F}\mathbf{P}\mathbf{S} - \mathbf{S}\mathbf{P}\mathbf{F} \quad (4)$$

vanishes.

4. Let ψ_g and ψ_u be orthonormal orbitals and define

$$\psi_\alpha(\theta) = \cos \theta \psi_g + \sin \theta \psi_u, \quad (5)$$

$$\psi_\beta(\theta) = \cos \theta \psi_g - \sin \theta \psi_u, \quad (6)$$

with $0 \leq \theta \leq \pi/4$. Using $\langle \hat{S}^2 \rangle = 1 - |\langle \psi_\alpha | \psi_\beta \rangle|^2$, derive the spin expectation value as a function of θ and explain the spin character of the two endpoint determinants.

5. For an infinitesimal rotation between an occupied orbital i and a virtual orbital a , suppose

$$\left. \frac{dE}{d\kappa} \right|_{\kappa=0} = 4F_{ia}. \quad (7)$$

Use variational stationarity together with the Slater–Condon relation

$$\langle \Phi_i^a | \hat{H} | \Phi_{\text{HF}} \rangle = F_{ai} \quad (8)$$

to establish Brillouin’s theorem.

Topic 2: Density Functional Theory

Problem 2: DFT Theorems, Potentials, and Exact Conditions

The following parts develop central DFT results through short symbolic derivations rather than numerical substitution.

1. Suppose two external potentials $v(\mathbf{r})$ and $v'(\mathbf{r})$, which do not differ only by a constant, have distinct nondegenerate ground states but the same ground-state density. Prove by applying the strict Rayleigh–Ritz inequality in both directions that this assumption leads to a contradiction, thereby establishing the first Hohenberg–Kohn theorem for this case.
2. Starting from the Hartree functional

$$E_{\text{H}}[n] = \frac{1}{2} \iint \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}', \quad (9)$$

derive its functional derivative $v_{\text{H}}(\mathbf{r})$.

3. Define the uniformly scaled density $n_{\gamma}(\mathbf{r}) = \gamma^3 n(\gamma\mathbf{r})$ and assume a local exchange functional

$$E_x[n] = -C \int n(\mathbf{r})^p d\mathbf{r}. \quad (10)$$

Use the exact scaling condition $E_x[n_{\gamma}] = \gamma E_x[n]$ to determine the exponent p .

4. For the grid LDA exchange energy

$$E_x^{\text{grid}} = -C_x \sum_k w_k n_k^{4/3}, \quad (11)$$

derive its first-order change under small density errors δn_k and use the result to explain why exact conservation of the grid electron number does not guarantee an accurate exchange energy.

5. For a normalized one-electron density, show that exact exchange cancels the Hartree self-repulsion and that the exact correlation energy vanishes, thereby deriving the one-electron condition

$$E_{\text{H}}[n] + E_x[n] + E_c[n] = 0. \quad (12)$$

Topic 3: Configuration Interaction and Coupled-Cluster Theory

Problem 3: Core Ideas in Configuration Interaction and Coupled-Cluster Theory

The following five models review determinant spaces, Hamiltonian structure, correlation through configuration mixing, the coupled-cluster exponential, and size consistency. Use spin-orbital excitation operators unless stated otherwise.

1. Consider four electrons in four spatial orbitals with $N_\alpha = N_\beta = 2$ and Hartree–Fock reference

$$|\Psi_{\text{HF}}\rangle = |1\bar{1}2\bar{2}\rangle, \quad (13)$$

where spatial orbitals 1 and 2 are occupied. Construct the excitation-rank count from the reference through quadruple excitations and use it to determine the dimensions of the CISD and full $M_S = 0$ CI spaces.

2. Order the CISD basis as the reference, all single excitations, and all double excitations. For canonical Hartree–Fock orbitals and an electronic Hamiltonian containing only one- and two-electron operators, derive the CISD Hamiltonian in block-matrix form, marking every identically zero block and justifying the structure with the Slater–Condon rules and Brillouin’s theorem.
3. In the basis $\{|\Psi_{\text{HF}}\rangle, |\Psi_D\rangle\}$, consider

$$\mathbf{H} = \begin{bmatrix} E_{\text{HF}} & V \\ V & E_D \end{bmatrix}, \quad \Delta_D = E_D - E_{\text{HF}} > 0. \quad (14)$$

In the weak-coupling limit $|V| \ll \Delta_D$, derive the ground-state energy through second order in V and the ratio of the double-excitation coefficient to the Hartree–Fock coefficient through first order, then interpret the signs and the double-excitation weight.

4. Consider the CCSD wavefunction

$$|\Psi_{\text{CCSD}}\rangle = e^{\hat{T}_1 + \hat{T}_2} |\Psi_{\text{HF}}\rangle, \quad (15)$$

with cluster operators

$$\hat{T}_1 = \sum_{ia} t_i^a a_a^\dagger a_i, \quad (16)$$

$$\hat{T}_2 = \frac{1}{4} \sum_{ijab} t_{ij}^{ab} a_a^\dagger a_b^\dagger a_j a_i, \quad (17)$$

use the conventional antisymmetric spin-orbital amplitudes to derive the CI single and double coefficients generated by the exponential, and explain the disconnected antisymmetrized product appearing in the double coefficient.

5. Two infinitely separated fragments have

$$|\Psi_X\rangle = (1 + c_X \hat{\tau}_X) |\Phi_X\rangle, \quad X = A, B, \quad (18)$$

where $\hat{\tau}_X$ is a double-excitation operator localized on fragment X . Demonstrate from the fragment product and $e^{t_A \hat{\tau}_A + t_B \hat{\tau}_B}$ why CCD is size-consistent while CID is not, explicitly identifying the disconnected quadruple and the additive energy for $\hat{H} = \hat{H}_A + \hat{H}_B$.

Topic 4: Excited States

Problem 4: Core Models for Molecular Excited States

The following five models emphasize spin adaptation, response theory, single–double mixing, optical intensity, and method selection.

1. Let $|\Phi_\alpha\rangle = |u\bar{g}\rangle$ and $|\Phi_\beta\rangle = |g\bar{u}\rangle$ be the two $M_S = 0$ singly excited determinants of a two-electron closed-shell molecule. Choose their phases so that the symmetric combination is the singlet, and consider

$$\mathbf{A} = \begin{bmatrix} \Delta + K - J & K \\ K & \Delta + K - J \end{bmatrix}. \quad (19)$$

Diagonalize this CIS matrix, identify the singlet and triplet roots, and derive their energy splitting.

2. For one dominant occupied-to-virtual transition, the TDHF eigenvalue problem is

$$\begin{bmatrix} A & B \\ -B & -A \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = \omega \begin{bmatrix} X \\ Y \end{bmatrix}. \quad (20)$$

Derive the positive TDHF root and the CIS/Tamm–Dancoff result, obtain the weak- B/A expansion for $A > 0$, and interpret the limits in which the root vanishes or becomes imaginary.

3. Model either a local or charge-transfer excited state $X \in \{\text{local}, \text{CT}\}$ by coupling a singly excited configuration to a higher double excitation:

$$\mathbf{H}_X = \begin{bmatrix} E_S^X & V_X \\ V_X & E_D^X \end{bmatrix}, \quad \Delta_X = E_D^X - E_S^X > 0. \quad (21)$$

Derive the weak-coupling energy correction and double-to-single coefficient ratio, and use $\lambda_X = |V_X|/\Delta_X$ to explain why CIS is less reliable when $\lambda_{\text{CT}} \gg \lambda_{\text{local}}$.

4. Two CIS configurations have x -directed transition-dipole matrix elements d and $-d$, with $d \neq 0$, and form the normalized coefficient vectors

$$\mathbf{c}^{(1)} = \frac{1}{\sqrt{2}}(1, 1), \quad \mathbf{c}^{(2)} = \frac{1}{\sqrt{2}}(1, -1). \quad (22)$$

For real coefficients, the transition dipole and oscillator strength are

$$\mu_x^{(n)} = - \sum_{k=1}^2 c_k^{(n)} d_k, \quad f_n = \frac{2}{3} \omega |\boldsymbol{\mu}_n|^2. \quad (23)$$

Assuming both states have the same positive excitation energy ω , derive their transition dipoles and oscillator strengths, identify the bright and dark combinations, and explain the role of relative phase.

5. An excited state is dominated by a double excitation relative to the ground-state determinant. Assess the suitability of CIS, adiabatic TDHF or TDDFT, Δ SCF, and EOM-CCSD for this state, then identify a higher-level or multireference approach appropriate for benchmark accuracy.

Topic 5: Quantum Algorithm Basics for Quantum Chemistry

Problem 5: From Occupation Qubits to Molecular Energies

This problem follows the main steps of a small quantum-chemistry calculation: encoding occupations, preparing a symmetry-preserving state, mapping a fermionic operator to Pauli strings, minimizing a variational energy, and resolving an energy by phase estimation. We write occupation states as $|n_0 n_1 \cdots n_{M-1}\rangle$, with spin orbital 0 stored in the leftmost bit.

1. The occupation and total-number operators are

$$\hat{n}_p = \frac{I - Z_p}{2}, \quad \hat{N} = \sum_{p=0}^{M-1} \hat{n}_p. \quad (24)$$

Show that an arbitrary occupation string is an eigenstate of $\hat{N}$ with eigenvalue $\sum_p n_p$, apply the result to the minimal-basis H_2 RHF determinant $|1100\rangle$, and explain why particle-number-preserving ansatzes avoid symmetry leakage.

2. Starting from $|00\rangle$, apply X to q_0 , H to q_1 , and then $\text{CNOT}_{1 \rightarrow 0}$. Use $X|0\rangle = |1\rangle$, $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, and the convention that this CNOT flips q_0 only when $q_1 = 1$. Derive the final state and show that it is entangled while remaining an eigenstate of $\hat{N} = \hat{n}_0 + \hat{n}_1$. You may test entanglement by arranging the amplitudes $C_{q_0 q_1}$ into a 2×2 coefficient matrix: a product state has a rank-one matrix.
3. For two fermionic modes, use the Jordan–Wigner operators

$$a_0 = \frac{1}{2}(X_0 + iY_0), \quad a_0^\dagger = \frac{1}{2}(X_0 - iY_0), \quad (25)$$

$$a_1 = \frac{1}{2}Z_0(X_1 + iY_1), \quad a_1^\dagger = \frac{1}{2}Z_0(X_1 - iY_1). \quad (26)$$

Derive the Pauli representation of the hopping operator $a_0^\dagger a_1 + a_1^\dagger a_0$ and use its action on occupation states to explain why it conserves particle number.

4. In the one-electron sector, consider the real Hamiltonian

$$\hat{H}_{\text{eff}} = \bar{E}I + \frac{\delta}{2}(Z_1 - Z_0) + g(X_0 X_1 + Y_0 Y_1) \quad (27)$$

and the ansatz

$$|\psi(\theta, \varphi)\rangle = \cos \theta |10\rangle + e^{i\varphi} \sin \theta |01\rangle, \quad 0 \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq \varphi < 2\pi. \quad (28)$$

Derive $E(\theta, \varphi)$ and minimize it analytically for general real δ and g , including the limits $g = 0$ and $\delta = 0$.

5. Let $U(t) = e^{-i\hat{H}t}$ with $t > 0$ and $e^{-iE_k t} = e^{2\pi i \phi_k}$ for an exact eigenstate of energy E_k . Ideal m -bit phase estimation returns the unsigned integer y with $\phi_k = y/2^m$, and its energy resolution is approximately

$$\delta E \simeq \frac{2\pi}{t2^m}. \quad (29)$$

Derive the energy represented by y , including its integer phase-alias branch, and obtain the minimum m required for a target resolution ϵ_E . Explain the resulting tradeoff between increasing t and increasing m .